Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.23	20.05	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.00	-0.9%
power (W), mean	628	641	+2.0%
power (W), peak	651	686	+5.3%
energy (Wh)	212	214	+1.1%
energy, parked workers still on at idle power (Wh)	212	214	+1.1%
CPU used (cores), mean	0.874	1.21	+38.8%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.036	0.051	+38.8%
energy per core-hour (Wh)	719	529	-26.5%
node-hours per core-hour	6.87	4.95	-28.0%
pending pods, pod-minutes	0.000	0.000	0
pending pods, peak	0.000	0.000	0
HPA replicas, mean	8.89	9.26	+4.1%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.500	0.783	+56.7%
requests timed	1,035	1,100	+6.3%
time over the response line (% of samples)	0.773	0.000	-100.0%
response time (ms), mean	161	94.85	-41.1%
response time (ms), median	132	80.95	-38.4%
response time (ms), 95th percentile	325	154	-52.7%
response time (ms), 99th percentile	495	191	-61.5%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.4803	5.5020	-26.4%	[-3.6701, -0.2150]	GAIN
kwh_per_core_hour	0.7802	0.5838	-25.2%	[-0.3602, -0.0255]	GAIN
utilisation	0.0360	0.0514	+43.0%	[+0.0023, +0.0302]	GAIN
pending_pod_minutes_per_hour	0.0000	0.0000	n/a	[+0.0000, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	1.4819	2.4734	+66.9%	[-2.2228, +5.0711]	STEADY, no significant difference

What Omni-Compass did

- decisions: 120 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 4
- power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.327 0.333 0.34 0.346 0.353 0.359 0.365 0.37 0.376 0.381 0.387 0.392 0.397 0.402 0.407 0.412 0.416 0.421 0.425 0.429 0.434 0.438 0.442 0.446 0.45 0.454 0.458 0.462 0.465 0.469 0.472 0.476 0.479 0.482 0.485 0.488 0.491 0.494 0.497 0.5 0.502 0.505 0.508 0.51 0.513 0.516 0.518 0.521 0.523 0.525 0.528 0.53 0.532 0.534 0.537 0.539 0.541 0.542 0.544 0.546 0.548 0.55 0.552 0.553 0.555 0.556 0.557 0.558 0.559 0.56 0.561 0.562 0.563 0.564 0.564 0.565 0.566 0.567 0.568 0.568 0.569 0.57 0.571 0.572 0.572 0.573 0.574 0.575 0.576 0.576 0.577 0.578 0.579 0.579 0.58 0.581 0.582 0.582 0.583 0.583 0.584 0.585 0.585 0.586 0.586 0.586 0.587 0.587 0.587 0.588 0.588 0.588 0.588 0.588 0.589 0.589 0.589 0.589 0.59 0.59
- nodes decided per minute: 6 6 6 6 6 6 6 6 6 6 6 5 5 5 5 5 5 4 4 4 4 4 4 4 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	618	5.00	5.00	0.000	0.000	0.576	0.597
1	6.00	6.00	619	622	5.00	6.50	0.000	0.000	0.621	0.722
2	6.00	6.00	618	624	5.00	7.00	0.000	0.000	0.580	0.747
3	6.00	6.00	622	630	5.75	7.25	0.000	0.000	0.698	0.938
4	6.00	6.00	632	651	9.00	10.00	0.000	0.000	0.999	1.39
5	6.00	6.00	634	645	9.00	10.00	0.000	0.000	0.998	1.34
6	6.00	6.00	633	650	9.50	10.00	0.000	0.000	1.01	1.45
7	6.00	6.00	646	671	10.00	10.00	0.000	0.000	1.32	2.03
8	6.00	6.00	646	673	10.00	10.00	0.000	0.000	1.34	2.06
9	6.00	6.00	646	676	10.00	10.00	0.000	0.000	1.34	2.19
10	6.00	6.00	640	666	10.00	10.00	0.000	0.000	1.19	1.88
11	6.00	6.00	616	622	10.00	10.00	0.000	0.000	0.587	0.716
12	6.00	6.00	619	624	10.00	10.00	0.000	0.000	0.618	0.770
13	6.00	6.00	624	634	10.00	10.00	0.000	0.000	0.776	0.996
14	6.00	6.00	631	644	10.00	10.00	0.000	0.000	0.954	1.27
15	6.00	6.00	632	646	10.00	10.00	0.000	0.000	0.964	1.32
16	6.00	6.00	633	649	10.00	10.00	0.000	0.000	0.983	1.40
17	6.00	6.00	628	635	10.00	10.00	0.000	0.000	0.863	1.06
18	6.00	6.00	620	622	10.00	10.00	0.000	0.000	0.634	0.703
19	6.00	6.00	617	625	10.00	10.00	0.000	0.000	0.580	0.808

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.